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The role of contextual and individual factors in male mate choice for size in a marine snail

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Echinolittorina malaccana, a marine gastropod living on rocky shores along the Indo-West Pacific coasts, shows consistent mate choice in which males prefer to mate with females slightly larger than themselves. Previous studies suggest that the strength of this preference is, to a certain extent, context dependent, being influenced by both demographic and individual (potentially genetic) factors. To disentangle how exactly these factors contribute to male mate choice for size in *E. malaccana*, we employed modified versions of a previous mate choice experiment using a range of female and male size classes. Mate choice was measured from the same male individuals across different experimental days, thus allowing the analysis of the repeatability of male preferences, as well as the comparison of the strength of mate choice with previous experimental designs and data from wild populations. Two parallel experiments (presenting either the same or different female individuals to males across days) were conducted, which allowed the inference of whether size alone, or additionally with other traits, influences mate choice. Results showed that male preference for females slightly larger than themselves remained consistent in both experiments, although its specific characteristics (e.g. strength of preference, extent of size bias) differed from previous studies. This suggests that the estimation of mate choice in *E. malaccana* is sensitive to experimental approaches and, to a certain extent, context dependent. The repeatability of mate choice (0–0.36) was only significant in the experiments where males were presented with the same female individuals. Mate choice is, therefore, primarily based on size, but other factors such as female quality or imprinting may modify this choice and lead to discrepancies in repeatability between experiments.

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Mate choice, or the preference to mate with individuals bearing beneficial traits, is a widespread behaviour in most sexually reproducing organisms (e.g. Roff, 2015; Rosenthal, 2017). This trait is complex, involving interactions between a chooser (the animal that actively performs the choice) and a partner (the chosen individual; Dougherty, 2023). Given its significance to evolutionary processes, behavioural ecologists and evolutionary biologists have devoted a lot of effort to studying this phenomenon (Andersson, 1994). The most common way to assess mate choice in the wild has been to measure patterns within a population, such as positive and negative assortative mating (Dougherty, 2023; Jiang et al., 2013; Moura & Gonzaga, 2017). This approach, however, provides limited information, as causes other than mate choice can contribute to assortative mating (Andersson, 1994; Jennions et al., 1995; Rolán-Alvarez, Carvajal-Rodríguez, et al., 2015; Shine et al., 2001; Taborsky et al., 2009) and, therefore, laboratory experiments have been used to complement natural observations (Harari et al., 1999; Li et al., 2022; McLain et al., 2015; Moura & Gonzaga, 2017; Rueger et al., 2016).

While mate choice studies have targeted various traits, size has been the most frequently studied (Dougherty, 2023; Rios Moura et al., 2021; Rosenthal, 2017). Size-assortative mating has been detected in nearly all animal groups (Dougherty, 2023; Jiang et al., 2013; Rios Moura et al., 2021; Rosenthal, 2017), including humans (Stulp et al., 2017). Sexual size dimorphism is, for example, widespread in marine gastropods, and mating pairs are typically composed of a female slightly larger than the male (Ng et al., 2019). Such male–female size difference in mating pairs has been explained by a mechanism of mate choice whereby males tend to choose females slightly larger than themselves, resulting in a positive assortative mating for size and strong sexual selection in females (Ng et al., 2019).

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Such a positive-size assortative mating pattern has also been reported in *Echinolittorina malaccana*, a species of snail living in the upper intertidal areas of the Indo-West Pacific region which has a planktotrophic developmental stage (Lau et al., 2021; Ng et al., 2019). *Echinolittorina malaccana* shows an overall positive relationship between size and both age and fecundity (Mak, 1996; Lau et al., n.d.), an association that has been shown in other species of the same family (with direct development though; reviewed in Reid, 1996; Rolán-Alvarez, Boulding, & Austing, 2015). *Echinolittorina malaccana* also exhibits seasonal reproduction with mating occurring in spring and summer (Ng et al., 2019). During the mating season, snail activity follows the tidal cycle, with individuals mating when awash (predominantly on rising tides), and entering an aestivation state when emersed (Ng et al., 2021). Males show an active mate choice for females slightly larger than themselves via mucus trail following (Lau et al., 2021; Ng et al., 2021). They typically mount and circle on the chosen female, until finally settling in the copulatory position where they insert the penis into the female cavity and initiate copulation (Ng et al., 2021). Specifically, males display, on average, a preference for females that are 6–18% larger than themselves (Lau et al., 2021; Ng et al., 2019). Females, in contrast, exhibit limited choice of their mating partners, showing little reaction to copulation and continuing to forage (Ng, Davies, et al., 2016, 2019). Such differences in mating behaviours between the sexes, common in marine gastropods, are thought to be caused by females having sperm reservoirs that can store sperm from multiple males for months (i.e. they have a reduced need to mate continuously), while males attempt to mate with as many females as possible during the mating season (Ng et al., 2019). Despite the overall consistent pattern, the strength of the male preference and the optimum female size chosen vary depending on natural versus laboratory settings and the experimental design used to study them. The strength of the male preference, for example, decreases from a laboratory multiple-choice design to measurements on the shore, followed by male-choice and finally single-choice designs (see Lau et al., 2021).

While mate choice has been well characterized in this species at a population level, these estimates do not fully account for how individual differences and contextual factors (demography, snail abundance, subtle changes in the experimental designs used, etc.) interact to contribute to the overall pattern observed. Intuitively, as in most traits, there should be a genetic basis behind individual differences in mate choice (Rosenthal, 2017) as this allows such a trait to evolve, but heritability has been shown to be low in many fitness-related traits (Caballero, 2020). Consequently, to further understand the evolution of mate choice, assessment at an individual level and estimation of a heritability proxy are needed. Broad (or even strict) sense heritability cannot be estimated in this species, as its planktonic life stage makes it challenging to rear in laboratory conditions. It is, however, possible to estimate an upper limit of the broad sense heritability, that is, repeatability (Wolak et al., 2012). Repeatability (R) effectively measures the proportion of total phenotypic variance in a population that is attributed to consistent between-individual variation (Sokal & Rohlf, 1995). In terms of variance components, repeatability can be computed by dividing the sum of genetic (V_G) and general environmental (V_{EG}; i.e. stable environmental factors that influence individual variation) variances by the overall phenotypic variance (V_P; i.e. $R = (V_G + V_{EG}) / V_P$; Caballero, 2020). Repeatability constitutes, therefore, an upper limit for the broad sense heritability (H^2), and provides an estimate of the potential role of genetic variation in mate choice, a requisite for the evolution of this trait. The repeatability for assortative mating, however, has been found to be rather low across several taxa (Rios Moura et al., 2021).

In this study, we carried out two laboratory experiments in which males of different sizes (and so most likely ages) were allowed to mate with females of five different size classes using a male-choice design (i.e. one male with five females) that was repeated over a series of days. In one experiment we presented different sets of female individuals of various size classes to the males during the experimental period (i.e. testing whether size is the primary determinant of male mate choice) and in the other we presented the same set of female individuals of various size classes on each day (i.e. testing whether individual-based factors, in addition to size, influence male mate choice). By comparing the male mate choice and the corresponding repeatability in these parallel experiments, the contribution of size and other individual-based factors in the females to the male mate choice, and its potential genetic basis, can be elucidated. Thus, while the first objective of this study was to confirm how changes in the experimental design (i.e. contextual factors) may affect the male mate choice for size displayed by *E. malaccana*, the second objective was to provide insights into how individual factors can influence the pattern of repeatability. If males in both experiments behaved in the same manner, a genetic preference for size alone could be the main determinant of choice, whereas, conversely, if there were marked differences in the behaviour of the males across experiments, factors other than a genetic preference for size (i.e. preferences for specific female individuals) could also be relevant to mate choice.

METHODS

Snail Collection and Maintenance

Snails ($N \sim 2000$, see Ethical Note) were collected from the Cape d' Aguilar Marine Reserve (22°12′33″N, 114°15′28″E) on 7 and 8 August 2023 (the peak mating season is from June to September in Hong Kong; Mak, 1998; Ng, Davies, et al., 2016). Individuals were transported to the laboratory at The Swire Institute of Marine Science, measured (shell length ± 0.1 mm with a vernier calliper), and sexed (determined by the presence/absence of a penis). Males were identified by a small paint dot on the shell (which has minimal effects on their behaviours; S. Lau, personal observation). All the snails, irrespective of the experiment in which they were later involved, were kept in air under dry conditions to imitate low tide situations until they were used in the experiments (conducted on 5 separate days: 9–11, 14–15 Aug 2023). This protocol mimics the natural behavioural cycle on the shore, where snails remain inactive when emersed during low tides, then become active and start to mate when wetted by the rising tides (Ng et al., 2019, 2021). Snails were maintained in air in the laboratory for 2–8 days (except during feeding; see below), which is within the range of emersion durations as this high shore species can experience days and even weeks without tidal inundation during neap tides (Marshall et al., 2013). Snails were fed with a diluted phytoplankton solution on which they graze (Seachem, Madison, GA, U.S.A.; S. Lau, personal observation) for 1 h on 12 Aug 2023. After feeding, all snails were blotted dry and maintained in air until they were used in the experiments. After the experiments, all snails were returned to the shore alive (see Ethical Note).

Laboratory Mate Choice Experiments for Size

Following a previous study on mate choice in this species (Lau et al., 2021), mating chambers (transparent plastic spheres, 12 cm diameter) were used in two parallel experiments to observe snail matings using a male-choice design, where one male is transferred to each mating chamber and presented with multiple females of different sizes (Fig. 1a). This design is more efficient than a single-choice design (one male and one female per chamber) to achieve sufficient mating observations within a limited time to measure

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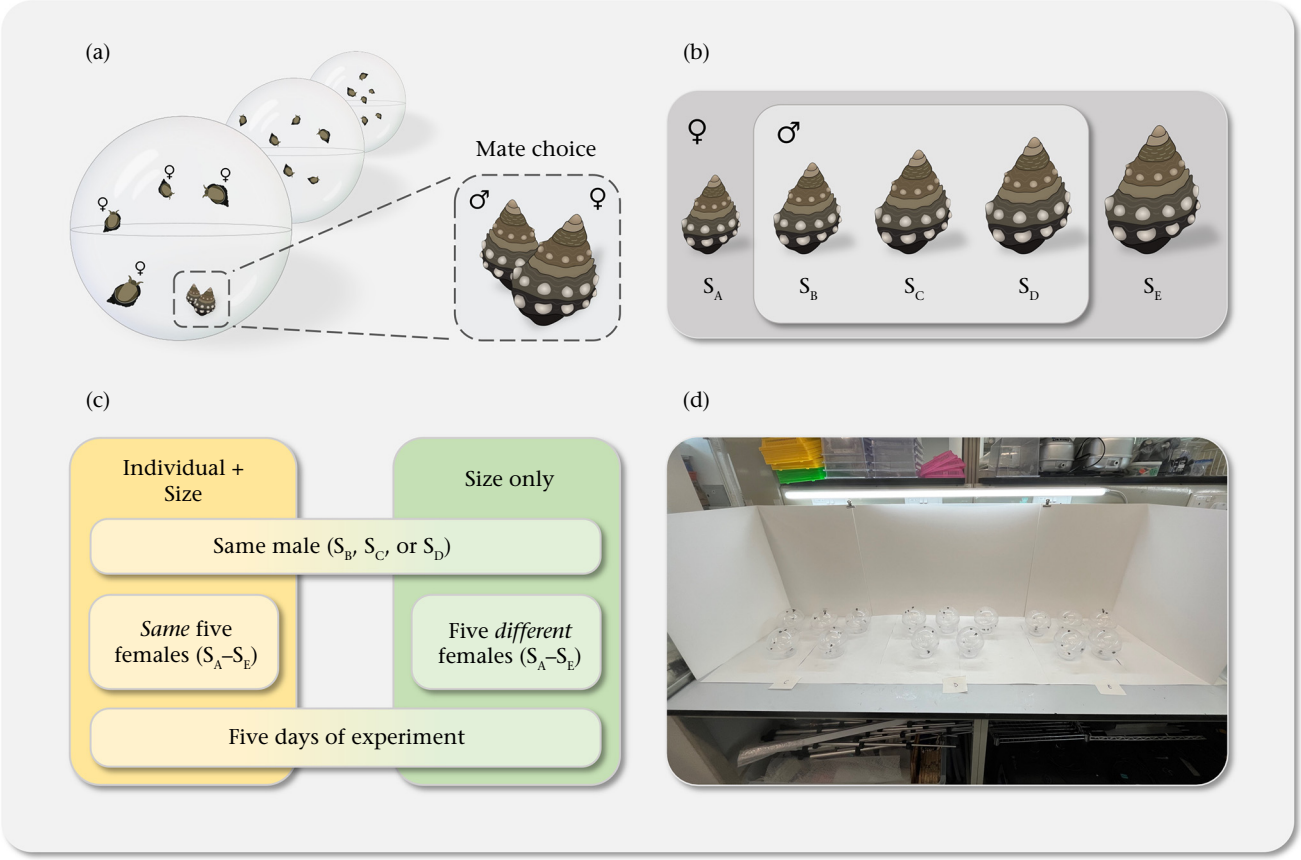


Figure 1. Details of the experimental design. (a) Mating chambers (transparent plastic spheres) were used to study the mate choice-driven assortative mating for size in *E. malaccana*. In a male-choice design, a male could choose to mate with any of the five females of different sizes inside the chamber, and the first mating (where the male mounted the female and remained in the copulatory position) was recorded. (b) Five female size classes ($S_A - S_E$) and three male size classes ($S_B - S_D$) were used in the experiments (10 replicates for each male size class). S_A: 6–6.9 mm; S_B: 7–7.9 mm; S_C: 8–8.9 mm; S_D: 9–9.9 mm; S_E: 10–10.9 mm. See text for further details. (c) Two parallel experiments were run to test whether female size or other individual factors contribute to male mate choice: 'Individual + Size' (I+S) and 'Size Only' (SO). In each experiment, the same male individuals (of classes S_B, S_C, or S_D) were used per chamber. In contrast, the females presented to the males in the chambers were either the same set of individuals (I+S experiment; classes $S_A - S_E$), or different sets of individuals of equivalent size classes (SO experiment). Both experiments lasted 5 days (the first 3 days were consecutive, followed by a break to feed the snails, and finally 2 consecutive days). (d) Laboratory se-tup during 1 day of the experiment.

repeatability. A male-choice design was also preferred over a multiple-choice design given its statistical independence for subsequent analyses.

Five size classes were used (S_A: 6–6.9 mm; S_B: 7–7.9 mm; S_C: 8–8.9 mm; S_D: 9–9.9 mm; S_E: 10–10.9 mm). Mate choice was examined in three male size classes ($S_B - S_D$) and each male was presented with five females of different sizes ($S_A - S_E$; Fig. 1b). The size classes in the present study differed from those in Lau et al. (2021), in which four nonoverlapping size classes were used (i.e. 4 – 4.9 mm, 6–6.9 mm, 8–8.9 mm and 10–10.9 mm), to avoid using small individuals (<6 mm), which have a low mating efficiency, and to cover the full distribution of size classes (i.e. without gaps between classes). With this modification in the experimental design, how the strength of mate choice varies with subtle changes in the methods (i.e. contextual factors) can be assessed by comparing the results with those of previous mate choice experiments and survey data from the wild. Using these size classes, we tested a size difference (based on average differences between size classes) in a mating pair of 3 mm maximum (as opposed to 6 mm in Lau et al., 2021), which is still sufficient to capture the male preference for females larger than themselves, as this ranges between 0.7 and 1.4 mm (see Lau et al., 2021).

To examine the repeatability (i.e. the genetic basis) of mate choice as well as to what extent this is influenced by preferences for particular female individuals, we assessed the male preference for specific female sizes by recording the choice of the same male individuals daily across 5 days of experiments. To test whether female size alone or other individual factors contribute to such preference, two parallel experiments were run: either the same set of female individuals (one individual from each of the $S_A - S_E$ size classes; denoted as 'Individual + Size' I+S experiment) or different sets of female individuals of equivalent size classes (denoted as 'Size Only' SO experiment; see Fig. 1c) were presented to the same male in a given mating chamber over the 5 days of the experiments. For each experiment, 10 replicate male individuals of three size classes ($S_B - S_D$) were examined (i.e. $\Sigma N = 10$ males $\times$ 3 size classes $\times$ 2 experiments $\times$ 5 days $=$ 300 separate mating chambers for observations in total for the entire experiments). This design allows us to partition the variation in the preferred female size into the between- and within-male components. A strong between-male component indicates differential preferences between males and, simultaneously, a high correlation between the preferred female size within males (i.e. repeatability). For procedural reasons, replicates were divided into two batches on each day, with five replicate males from each size class of each experiment observed in the morning and another five in the afternoon (a total of 30 mating chambers running simultaneously in each batch).

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All the individual snails used in both experiments were handled and treated equally. Before the experiments, all the mating chambers were sprayed with sea water to provide a moist environment, which induces snail activity. All individual snails were also wetted with sea water to stimulate foot extension before being transferred to the chambers. In each chamber, females of different sizes were sequentially added to a fixed starting point (in randomized order for each chamber on each day) and followed by the male so that the male was in contact with the mucus trails laid by all previously introduced females (see Lau et al., 2021). The experimental sessions were run for a maximum of 2 h, and once the first mating pair (defined as a male mounting a female and remaining in the copulatory position for more than 5 s; Ng et al., 2019; see Fig. 1a) was recorded, all snails were removed from the chamber to prevent further interactions. If snails were observed to aggregate and stop moving, the chambers were sprayed with sea water to stimulate movement. After the experimental sessions on each day, all snails, irrespective of the experiment, were blotted dry and maintained in air until another session on the next day. This process mimics what the snails experience in the wild, where they remain inactive during the low tide. Mating chambers were shielded with white cardboard sheets to minimize visual stimuli to the snails (see Fig. 1d) and scrubbed with 95% ethanol to remove mucus trails before rinsing and reusing.

Statistical Analyses

To investigate whether the various independent variables, including male size classes (fixed, three levels), female size classes (fixed, five levels), experiments (I+S or SO; fixed, two levels), experimental days (random, five levels) and batches (random, two levels), contributed to the frequency of observed mating, a logistic multiple regression analysis was performed which considered potential combinations of mating pairs in each chamber as either '1' for the recorded mating pair or '0' for the remaining nonmated combinations (response variable; Sokal & Rohlf, 1995). This analysis revealed that only female size classes and experiments made a significant contribution to mating (see Results), allowing the pooling of data when necessary for the rest of the variables. In addition, the homogeneity of mating frequencies with the different female size classes ($S_A - S_E$) per combinations of various groups and conditions (including experiments, male size classes and days) was checked using chi-square homogeneity tests (Sokal & Rohlf, 1995) to ensure data can be pooled to improve the sample size for subsequent mate choice and sexual selection effect estimations. Pooled data were then used to assess the strength of sexual selection (G_{sexual}) and presence of mate choice (using differential size classes, i.e. $G_{Dif\text{-}size}$; see below) using G likelihood tests, and to estimate proportional differences in matings across female size classes by bootstrapping the observed frequencies of matings (after normalizing to the size class with the highest frequencies) using the software JMATING 1.0 (Carvajal-Rodriguez & Rolan-Alvarez, 2006).

To test for the presence of mate choice (i.e. male preference) for female size, the frequency data of mating pairs were analysed using the mean size class differences between the males and females in a mating pair (hereafter 'differential size classes'; following Lau et al., 2021). The 15 possible combinations of male and female size classes (5 female size classes x 3 male size classes) were converted into seven differential size classes (mean differences of −3, −2, −1, 0, 1, 2 and 3 mm between individuals involved in a mating, with positive differences indicating females larger than males and assuming an average difference of 1 mm between consecutive size classes). Deviations from random mating were then checked by comparing the observed frequency of differential size classes to its expected frequency assuming random mating (i.e. −3 = 1/15, −2 = 2/15, −1 = 3/15, 0 = 3/15, 1 = 3/15, 2 = 2/15, 3 = 1/15) using a goodness of fit G test ($G_{Dif\text{-}size}$). Significant deviations from random mating, alongside significance in the sexual selection tests (as indicated by G_{sexual}) would suggest that males show a preference for certain female size classes. Finally, the size distributions of female−male differences in mating pairs were used to estimate the preference optimum (i.e. averaged male preference for female size). By comparing the results of these analyses to previous data on male mate choice for size in *E. malaccana* obtained from surveys or laboratory experiments, we can derive information as to how mate choice varies with contextual factors, such as the size classes used in the experimental designs.

An ANOVA was used to partition the variance in mate choice (using the size of the chosen female/s as the dependent variable) within (i.e. across days) and between male individuals of a given size class, and the repeatability of mate choice (R) was estimated as the intraclass correlation coefficient (ICC) between these two variances (Sokal & Rohlf, 1995). This analysis was performed for each of the two experiments (S+I and SO) with either the three male size classes considered separately (i.e. six different groups, with a maximum $N = 50$), or pooling the male size classes within experiments (i.e. two experiment groups, with a maximum $N = 150$), as increased sample size can improve the statistical power to detect repeatability (Wolak et al., 2012). A significant ICC (i.e. high repeatability) in mate choice for size implies a higher variance in the size of the chosen females between different male individuals compared to intramale variance in female sizes. For the comparison between experiments, if a (presumably genetic) male preference for larger sizes alone was the major determinant of mate choice, similarly high and significant Rs should be found in both experiments. Conversely, if individual factors other than the preference for certain sizes (e.g. imprinting or other individual effects) influence the mate choice process, the I+S experiment should show higher Rs than the SO experiment in the male mate choice for size. All statistical analyses were conducted in Excel and SPSS 24.0 version (SPSS Inc., Chicago, IL, U.S.A.).

Ethical Note

Echinolittorina malaccana was sampled from a nonendangered and high-density population (>400 individuals/m^2) over a 60 m stretch of rocky shore in Cape d' Aguilar Marine Reserve, Hong Kong under permit No (12) in AF MPD 09/3 Pt.28 issued by the Agriculture, Fisheries and Conservation Department, HK SAR Government. Approximately 2000 individuals were collected, most of which were sexed and maintained in air under dry conditions, and 960 snails directly participated in the experiments (i.e. 60 males with 150 females in the I+S experiment and 750 females in the SO experiment; see Table A1). We tried to follow the ASAB guidelines for the ethical treatment of nonhuman animals used in behavioural studies (ASAB Ethical Committee/ABS Animal Care Committee, 2023) throughout the experiments. The maintenance conditions of the snails in the laboratory mimicked those they experience daily in the wild and all snails were returned alive to the same area of rocky shore after the experiments.

RESULTS

Overall, a total of 177 mating pairs were recorded throughout the experiments (i.e. a mating efficiency of 59%), with 79 in the I+S and 98 in the SO experiments (Table 1). Almost all males (58 of 60) mated at least once across the 5 days of the experiments, while 88% (53 males) mated more than once and were used in the repeatability estimation. Only experiments and female size classes (logistic regression: Wald = 3.7, $P = 0.054$ and Wald = 43.4, $P \leq 0.001$,

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Table 1
Analyses of sexual selection and mate choice in *E. malaccana*

Study	Experiment/Design	*N*	G_{sexual}	$G_{Dif\text{-}size}$
Present study	I+S	79	22.5***	18.2**
	SO	98	11.5*	17.9**
Lau et al. (2021)	Male choice[1]	48–49	34–59***	110–165***
	Multiple choice	35	46***	157***
	Wild data	228	52***	2147***

Results of G likelihood tests comparing the observed and expected mating frequencies for sexual selection (G_{sexual}) and mate choice for differential size classes ($G_{Dif\text{-}size}$). Experiments: I+S: Individual + Size; SO: Size Only. Results from a previous study (Lau et al., 2021) using the same species are also presented for comparison.
*$P \leq 0.05$; **$P \leq 0.01$; ***$P \leq 0.001$.
[1] Note that two slightly different methods of the male-choice design were used (see Lau et al., 2021).

respectively) contributed to whether a pair of snails mated or not, whereas male size classes, experimental days and batches had no effect ($P > 0.7$ in all cases). When the same analysis was repeated for each experiment (I+S or SO) separately, again only female size was included in the regression model ($P > 0.7$ for the rest of the variables). In the subsequent repeatability analyses, therefore, male size classes, days and batches were pooled, to focus on the effects of the experiments and female size classes on the resulting mating patterns (see below).

Mating frequencies involving the different female size classes were homogeneous across all groups (chi-square homogeneity tests), except when testing the male size class S_B across days in the I+S experiment (Table A2). A detailed inspection (i.e. comparing each day of the experiment against the pooled remaining days) revealed that this heterogeneity was caused by just the results from 1 day (day 5). Consequently, the mating frequencies were pooled across male size classes and days (excluding day 5 from SO) for the subsequent analyses of sexual selection and mate choice (Table 1, Fig. 2). Based on these results, male size classes were analysed separately and pooled within each experiment to increase sample size when estimating repeatability.

Sexual selection (G_{sexual}) was significant in both experiments (Table 1; although the distribution of size classes in mated females did not differ between experiments: $G = 6.5$, $P = 0.162$; Fig. 2a). When analysing mating frequencies with differential size classes ($G_{Dif\text{-}size}$), both experiments showed significant deviations from the expected frequencies assuming random mating, indicating a male-driven mate choice process (Table 1, Fig. 2b). Nonrandom mating was also detected from bootstrapping the observed mating frequencies (normalized to the size class S_E), where significant differences were detected in female size classes S_A and S_B of both experiments (Table A3). The size difference between the females and males from the mated pairs showed that, on average, females were 0.74 mm larger than the males in the I+S experiment, and 0.50 mm larger in the SO experiment, representing a size increase of 8.7% and 6% with respect to the average male size in the respective experiments.

The repeatability analyses detected significant results in two of three male size classes (S_B and S_D) in the I+S experiment, but not in any of the male size classes in the SO experiment (Table 2; excluding day 5 from data in SO). The average *R* values were 0.30 and 0.06 for the I+S and SO experiments, respectively (Table 2). A similar result was obtained when pooling all male size classes within experiments or when day 5 was included in the SO experiment, with significant repeatability detected only in the I+S but not the SO experiment.

DISCUSSION

As in most marine gastropods, *E. malaccana* shows both positive size-assortative mating and sexual selection for larger females as well as sexual size dimorphism (Ng et al., 2019). These processes may causally relate to male mate choice where, overall, males (the choosers) show a preference for females (the partners) that are larger than themselves (Lau et al., 2021; this study). The present study assessed whether mate choice for size in *E. malaccana* varies depending on contextual factors (such as when measured under different experimental designs, or when performing slight changes in the size classes used), and to what extent such strength is influenced by individual factors using repeatability (i.e. the I+S versus SO experiments).

Contextual Factors Influencing Mate Choice

Mate choice for size in *E. malaccana* appears to be a robust and widespread behavioural trait, as we found a similar pattern of mate choice to that described in the literature, with the females chosen being 8.7% and 6% larger than the males in the Individual + Size (I+S) and Size Only (SO) experiments, respectively, which is within the range estimated in previous studies (6% for single choice and 18% for multiple choice; Lau et al., 2021). The characteristics of this mate choice, however, appeared to vary across experimental

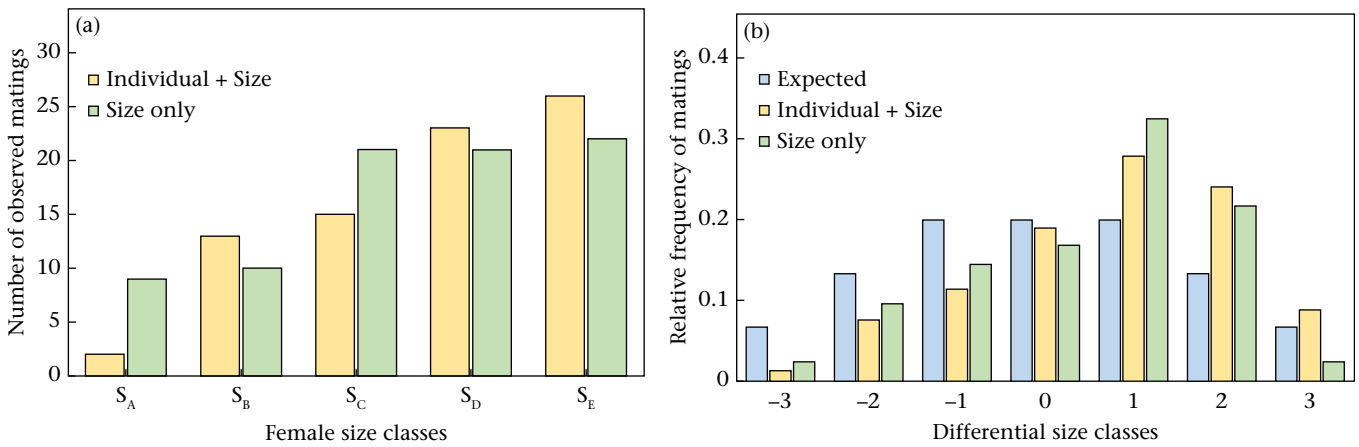


Figure 2. Matings for the 'Individual + Size' (I+S) and 'Size Only' (SO) experiments after pooling male size classes and days (excluding day 5 in SO). (a) Number of observed matings for different female size classes. S_A: 6–6.9 mm; S_B: 7–7.9 mm; S_C: 8–8.9 mm; S_D: 9–9.9 mm; S_E: 10–10.9 mm. (b) Relative frequencies of matings across differential size classes (i.e. the mean size class difference between the females and the males involved in a mating pair).

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Table 2
Repeatability estimates

Experiment	Male size class	R	P_{ANOVA}
I+S	S_B	0.51	0.015
	S_C	0.09	0.308
	S_D	0.41	0.038
	Average (± SE)	0.33 (± 0.220)	–
	Pooled	0.30	0.009
SO	S_B	0.21	0.167
	S_C	0.18	0.182
	S_D	−0.23	0.866
	Average (± SE)	0.05 (± 0.248)	–
	Pooled	0.06	0.284

Results of repeatability (R) analyses for the two experiments (I+S: Individual + Size; SO: Size Only), with male size classes (S_B: 7–7.9 mm; S_C: 8–8.9 mm; S_D: 9–9.9 mm) either considered separately, together with their average, or pooled. R is the repeatability estimate (ICC between males across days; see Methods) and P_{ANOVA} is the probability of the associated ANOVA test.

designs, highlighting that mate choice is sensitive to contextual and demographic conditions (e.g. the presence and relative frequency of both sexes and, more importantly, their distribution of ages and sizes). These factors seem to influence the mechanisms of mate choice and are manifested as a certain degree of plasticity (i.e. the same male may exhibit distinctive attitudes to different females of the same size class on different experimental days; see examples of mate choice plasticity in Qvarnström, 2001; Cotton et al., 2006; Rosenthal, 2017; Candolin, 2019). A similar influence of demographic and contextual factors has been observed, for instance, in the Bulwer's petrel, *Bulweria bulwerii*, in which assortative mating driven by mate choice was only observed in a large population from Selvagens archipelago but not in a small population from the Azores (Bried et al., 2021). Moura and Gonzaga (2017) also found that variation in mate choice and assortative mating in the spider *Manogea porracea* depended on sex ratio and population density. In wolf spiders, *Pardosa milvina*, mating success appears to depend not only on male premating condition (an honest indication of male quality) but also on experience (Hoefler et al., 2009). Finally, in a meta-analysis of mate choice in several animal groups, Dougherty (2023) found that the strength of mate choice depended on the size and parasite load of the chooser, but not the chooser's age, physical condition or mating status. It appears, therefore, that although the male mate choice for size remains a robust phenomenon in *E. malaccana* (and probably many other species), its characteristics (e.g. strength of preference, magnitude of size bias, and so on) depend on several interacting contextual factors.

In many gastropod species, males employ mucus trail following to locate and choose mates, from which they may acquire fitness-relevant information (such as the female's size and parasite load) before coming into any physical contact with the female (see Ng et al., 2013). In *E. malaccana*, this mate-assessing behaviour is supported by the observation that males trail-follow females specifically (the reverse is infrequent, particularly during rising tides) and most eventually mount and mate with the followed females (Ng et al., 2019). Such a mechanism of employing mucus trail following for reproductive purposes aids the efficiency of exhibiting mate choice in gastropods, particularly when there are low densities of conspecifics or there is a limited time window to mate (Ng et al., 2013). The common use of trail following by males to search for mates may, in part, contribute to the general mate choice for size pattern seen in many marine gastropod species (Ng et al., 2019). Such a shared trait may suggest a behavioural plesiomorphy (Ng et al., 2019) that is likely to confer advantages over random mating (Futuyma & Kirkpatrick, 2017; Rosenthal, 2017), which raises the question of what could be the specific fitness benefits of this behaviour. One hypothesis is that size-based male mate choice occurs because of the correlation between female size and fecundity (McLain et al., 2015; Rueger et al., 2016). The fitness advantage of mating with large females would, however, predict that males should prefer the largest females (McLain et al., 2015), not those only slightly larger than themselves, as observed in *E. malaccana* and other marine gastropods (Ng et al., 2019). An alternative explanation is that this size–fecundity correlation may operate in conjunction with a counteracting mechanism where females reject the smallest males (Baldauf et al., 2015). This, however, also seems unlikely in *E. malaccana*, as females do not seem to actively accept or reject any potential mates (Ng, Davies, et al., 2016, 2019; S. Lau, personal observation). Another possibility has been suggested by Shine et al. (2001), who pointed out that positive assortative mating is caused by a change in preference with ontogeny (i.e. old and young males exhibit different mating strategies given their different sperm storage abilities). The effect of age is not seen in *E. malaccana*, however, as preference did not seem to differ between male size classes (i.e. a proxy for age classes).

Other possibilities for the preference for slightly larger females (but not the largest) involve demographic factors. In terms of probability, the largest females are relatively rare in a population (they represent one of the tails of the size distribution), and so males have a low chance of finding and mating with these individuals. Consequently, a strong preference for the largest females may not confer a high fitness advantage in this polygamous species. Second, potential male–male competition (where larger males tend to outcompete smaller males in accessing the chosen mate; Ng, Davies, et al., 2016) may also deter males from preferring the largest females (particularly in small males, which are likely to lose in a competitive interaction), thus leading to the preference for females slightly larger than themselves. This preference might, therefore, constitute a trade-off between fecundity and the access to mates. This hypothesis is also supported by the fact that a consistent sexual size dimorphism (females being slightly larger than males) is found in *E. malaccana* and several other gastropod species, which may favour a preference for slightly larger females and so increasing the chance of finding a suitable partner.

Individual Factors Influencing Mate Choice

Apart from the effects of contextual factors on the strength of mate choice, this study also investigated the role of individual factors in male mate choice for size. Two of three male size classes (S_B and S_D) in the I+S experiment showed significant repeatability, but not the remaining size class (S_C). This inconsistency across size classes is likely to be an experimental artefact, as the estimation of repeatability in behavioural traits (mate choice in particular) can be inherently noisy (Dohm, 2002; Bell et al., 2009; Wolak et al., 2012), and the overall results (pooling all male size classes) also suggested significant repeatability (i.e. indicative of an actual phenomenon).

Overall, the repeatability of male mate choice estimated in the present study (range 0.09–0.51 and on average 0.30) suggests low to mid heritability, as would be expected from a trait that is observed in most marine gastropods and that has important fitness implications (such as fecundity and the chance of mating). Repeatability in mate choice for size has been assessed in at least four taxa (reviewed in Boake, 1989; Rosenthal, 2017): insects (Du et al., 1987; Fowler-Finn & Rodríguez, 2013; Hedrick, 1988; Hoeffler et al., 2009; Thornhill, 1983; Butlin and Hewitt, 1986); amphibians (Howard & Young, 1998); birds (Forstmeier & Birkhead, 2004) and teleost fishes (Aspbury & Basolo, 2002; Cummings & Mollaghan, 2006; Gabor et al., 2011; Godin & Auld, 2013; Howard et al., 1998; Hoysak & Godin, 2007; Lehtonen & Lindström, 2008; Li et al., 2022). In these studies, repeatability ranged from 0 to 0.82, which suggests not only that experimental conditions likely play a

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role in the variable outcomes obtained (Cummings & Mollaghan, 2006; Li et al., 2022), but also that there is species-specific variation in mate choice.

In contrast to the I+S experiment, none of the three male size classes in the SO experiment showed repeatability for mate choice (i.e. no repeatability was found when the females presented to the males on each of the experimental days were not the same specimens but individuals of equivalent sizes), despite exhibiting a similar mate choice pattern (i.e. males preferring females 8.7% and 6% larger than themselves in the I+S and SO experiments, respectively). The discrepancy between experiments suggests that, while female size is the trait that contributes the most to male mate choice in *E. malaccana* (explaining the similar mate choice pattern across experiments), other secondary individual-dependent factors might be taken into account by the male snails when choosing particular mates (hence the discrepancy in repeatability across experiments). For example, other characteristics with a direct or indirect implication for the health (and thus fitness) of the females (e.g. the degree of starvation, parasitism or thermal stress), although not directly tested in this study, may be manifested in the mucus, thus potentially influencing male mate choice. It is possible that the different sets of females used in the SO experiment showed substantial variation when it comes to these secondary factors, resulting in reduced consistency across days and thus lower repeatability. In addition, it has been suggested that sexual imprinting can condition future sexual behaviour in some species by fixing the first experience at earlier ages (Irwin & Price 1999). Males in the I+S experiment could, therefore, have somehow been imprinted by the female individuals with which they first mated, thereby raising the chances of mating with these specific females in the future. This mechanism, however, seems unlikely as males are mating on every awash period and, given the high-density populations (>400 individuals/m^2) they are living in, it is unlikely that they can recognize and access the same individuals repeatedly. A simple alternative explanation could be that our experimental estimate of size (shell height) is not the most representative in terms of how males assess female size. Males are thought to estimate female size by the width of the female's mucus trail (Ng et al., 2019), which may be more closely related to shell width than shell length. Under this explanation, males would choose mates based solely on size, but as estimated by mucus trail width rather than shell height. This and competing hypotheses need to be investigated in future studies.

Conclusions

The study of size-based mate choice in *E. malaccana* (Lau et al., 2021; López-Cortegano et al., 2020; Ng et al., 2016ab, 2019) allows insight into how mate choice might occur and could have evolved in males, leading to a better understanding of the sexual selection and assortative mating pattern observed. Mate choice seems to be exerted by the preference of individual males to choose females slightly larger than themselves, although the characteristics of such mate choice is sensitive to the conditions under which it is measured, highlighting the influence of contextual factors. Such male preference may have evolved as a trade-off between maximizing fecundity and at the same time the chance of finding mates. The results obtained regarding repeatability also suggest that male preference is primarily determined by female size, but other individual factors (such as female quality and the process of imprinting) could, secondarily, influence the patterns observed. Future studies to closely examine how the various mechanisms, both contextual and individual, interact and drive the resultant overall male preference for slightly larger females in *E. malaccana* would help construct a more holistic picture of the widely observed size-based male mate choice pattern in marine gastropods and other animals.

Author Contributions

E. Rolán-Alvarez: Writing – review & editing, Writing – original draft, Investigation, Formal analysis, Conceptualization. **Gray Williams:** Writing – review & editing, Supervision, Methodology, Conceptualization. **Juan Gefaell:** Writing – review & editing, Formal analysis, Data curation. **Sarah L.Y. Lau:** Writing – review & editing, Methodology, Investigation, Conceptualization.

Data Availability

The data set for this experiment is available in the Figshare repository: https://doi.org/10.6084/m9.figshare.24806457.v3.

Declaration of Interest

The authors declare that they have no conflict of interest.

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Appendix

Table A1
Sample characteristics

Size class	Parameter	I+S		SO	
		Male	Female	Male	Female
S_A	Size	–	6.4 ± 0.304	–	6.5 ± 0.289
	N		30		150
S_B	Size	7.4 ± 0.261	7.5 ± 0.269	7.5 ± 0.331	7.5 ± 0.279
	N	10	30	10	150
S_C	Size	8.5 ± 0.243	8.6 ± 0.339	8.3 ± 0.187	8.5 ± 0.287
	N	10	30	10	150
S_D	Size	9.3 ± 0.341	9.4 ± 0.283	9.3 ± 0.257	9.4 ± 0.271
	N	10	30	10	150
S_E	Size	–	10.4 ± 0.406	–	10.4 ± 0.344
	N		30		150

Size (mean ± SD) and number (*N*) of *E. malaccana* of different size classes (S_A: 6–6.9 mm; S_B: 7–7.9 mm; S_C: 8–8.9 mm; S_D: 9–9.9 mm; S_E: 10–10.9 mm) used in the experiments (I+S: Individual + Size; SO: Size Only). Note that no male snails of size classes S_A and S_E were used (see Methods for further details).

Table A2
Chi-square test results

Factors pooled	Groups		Comparisons	χ^2	*df*	*P*
No pooling	I+S experiment	Male size class S_B	Across days	14.06	12	0.297
		Male size class S_C	Across days	19.10	16	0.263
		Male size class S_D	Across days	22.32	16	0.133
	SO experiment	Male size class S_B	Across days	27.36	16	0.001
		Male size class S_C	Across days	17.43	16	0.358
		Male size class S_D	Across days	16.99	16	0.386
Days	Male size class S_B		Female size classes vs experiment	4.98	4	0.289
	Male size class S_C		Female size classes vs experiment	2.14	4	0.709
	Male size class S_D		Female size classes vs experiment	0.86	4	0.931
Days & male size classes	–		Female size classes vs experiment	4.66	4	0.324
Days & treatments	–		Female size classes vs male size classes	5.27	8	0.728

Chi-square homogeneity tests to check for homogeneity in mating frequencies with different female size classes (S_B: 7–7.9 mm; S_C: 8–8.9 mm; S_D: 9–9.9 mm) under different groups and conditions. I+S: Individual + Size; SO: Size Only. Note that, overall, data were homogeneous across all groups and conditions, except when comparing across days in the male size class S_B in the SO experiment.

Table A3
Nonrandom mating obtained from bootstrapping

Experiment	Female size classes				
	S_A	S_B	S_C	S_D	S_E
I+S	0.08*** ± 0.059	0.50* ± 0.183	0.58* ± 0.201	0.88 ± 0.277	1
SO	0.41** ± 0.175	0.45* ± 0.187	0.95 ± 0.321	0.95 ± 0.328	1

Proportional differences in matings for the different female size classes (S_A: 6–6.9 mm; S_B: 7–7.9 mm; S_C: 8–8.9 mm; S_D: 9–9.9 mm; S_E: 10–10.9 mm), as determined by bootstrapping the observed mating frequencies in the two experiments (I+S: Individual + Size; SO: Size Only). Note that the frequencies are normalized to size class S_E, which has the highest mating frequencies.
$P \leq 0.05$; $^{**}P \leq 0.01$; $^{***}P \leq 0.001$.

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